

UK System Sell Price (SSP) Forecasting — One-Page Summary

Companion to `docs/technical-report.md` · all figures sourced from that report

Objective. Probabilistic day-ahead + intraday forecast of the UK System Sell Price across all 48 half-hourly settlement periods (SPs), with calibrated 80% prediction intervals (PIs).

£28.96 WF MAE · STATICBASE	£27.68 WF MAE · KALMAN	38.0→79.8% 80% PI COVERAGE	£32.03 7-DAY HOLDOUT MAE	0.4275 SHAPE CORR (MEAN)
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MODEL — TWO-STAGE LEVEL-SHAPE HGBR

- Stage 1 (level):** daily price level as quantiles P10/P50/P90 — pinball-loss HistGradientBoostingRegressor, 84 features, CPIH-deflated targets.
- Stage 2 (shape):** per-SP deviation from the daily level (separate H+1 and H+2 heads). Day-ahead SP forecast = `q50_level` + `q50_shape`; bands from level quantiles + per-SP conformal widening.
- Training:** weekly age-gated retrain on a rolling 3-year, CPIH-adjusted window; strict zero-leakage — every feature references data before day D begins, valid for all 48 day-ahead SPs.

INTRADAY CORRECTION — EVERY 30 MIN

Inference-only (frozen HGBR): fetch newly-settled SPs from Elexon BMRS → splice actuals into the live curve → scalar Kalman level-bias $\hat{x}$ (random-walk prior). Correction decays with horizon: $\text{correction}(h) = \hat{x} \cdot \gamma^h$, $\gamma = 0.966/\text{SP}$ (halves by $\approx \text{SP}+20$). Uncertainty propagates to the P10–P90 bands via $\pm 1.28 \cdot \sqrt{P} \cdot \gamma^h$ — widening when few SPs have settled, shrinking as posterior variance P falls. Kalman adds **+4.5 pp** PI coverage over the static base.

CALIBRATED UNCERTAINTY — SPLIT-CONFORMAL PER-SP Δ

Per settlement position, $\delta(\text{sp}) = 80\text{th-percentile conformity score over the calibration set}$; bands widened by $\delta(\text{sp})$. Lifts 80%-band coverage **37.99% → 79.82%** in-sample over the 119-day, 4-fold walk-forward (5,709 SP-rows) — a calibration gain, not a better point model. *Live DA coverage 66.0% over 30 spring/summer days (N=1,418), under active monitoring.*

SHORT-HORIZON NOWCAST — PERSISTENCE

SSP is near-AR(1) (ACF lag-1 = 0.828), so persistence ($\text{SP}[t] = \text{SP}[t-1]$) is the Bayes-optimal h+1 predictor. DA+Kalman adds no value through $\approx h+4$ and only overtakes persistence at $\approx h+4.5$ ($\approx 5\text{--}6\%$ better by h+5); an HGBR nowcast prototype was 17% worse than persistence at h+1 and was not shipped. Production serves persistence for h+1–h+3 with regime-asymmetric empirical 80% bands (live coverage 79.4–79.8%).

RESULTS

Metric	Value
WF MAE — StaticBase (119 days, 4 seasonal folds)	£28.96/MWh
WF MAE — deployed Kalman corrector	£27.68/MWh
7-day holdout MAE (RMSE £42.46, n=336)	£32.03/MWh
Stage-1 level MAE	£18.41/day
Stage-2 shape correlation (mean)	0.4275
Peak-timing MAE	6.71 SPs
80% PI coverage (in-sample WF)	37.99% → 79.82%

Seasonal WF MAE (Kalman): Summer £25.77 · Autumn £30.50 · Winter £20.27 · Spring £34.28. Baselines: naive t–48 £36.34 · seasonal naive t–336 £29.40 · rolling-mean-24h £26.78.

Status. Day-ahead 48-SP profile published early (~01:00 UTC) to a live Streamlit dashboard; intraday Kalman layer updates every 30 min. CI-tested (pytest on push). The early-publish pipeline is validating against the prior 12:30 UTC daily pipeline before full cut-over. **Live dashboard:** uk-system-price-forecast.streamlit.app